Linear Algebra — Lecture 6

Partitioned Matrices, LU Factorization, Computing an Inverse,
Solution Sets of Linear Systems

Friedberg–Insel–Spence, *Linear Algebra* 4e, §3.2–§3.4
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.2, §1.5, §2.2, §2.4–§2.5

Topics. Partitioned matrices and block multiplication, the column–row expansion of AB, inverses of block matrices, the LU factorization and solving a system with it, the algorithm for computing A^{-1}, the solution sets of homogeneous and nonhomogeneous systems, and reading a solution set off the reduced row echelon form.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$, while Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Partitioned matrices	§2.4 (p. 117–118)	—
Column–row expansion	§2.4 (p. 119)	—
Inverse of a block matrix	§2.4 (p. 119–120)	—
The LU factorization	§2.5 (p. 123–124)	—
Solving with LU	§2.5 (p. 124–125)	—
The LU algorithm	§2.5 (p. 125–127)	—
Computing an inverse	§2.2 (p. 107–108)	§3.2 (p. 161–163)
Invertibility and rank	§2.3 (p. 111–112)	§3.2 (p. 152–159)
Homogeneous systems	§1.5 (p. 43–47)	§3.3 (p. 171–172)
Nonhomogeneous systems	§1.5 (p. 43–47)	§3.3 (p. 172–174)
Reduced echelon form, basic variables	§1.2 (p. 12–21)	§3.4 (p. 185–190)

Two questions that drive the whole lecture: (1) *After a matrix is cut into blocks, do multiplication and inversion still work block by block?* (2) *Once the same matrix is written as $A = LU$, what does that save when solving, when inverting, and when counting operations?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Partitioned Matrices

The first five lectures followed one line: vector spaces, linear transformations, matrix representations. Every linear map became a matrix, a change of basis became Q on one side and Q^{-1} on the other, and the rank was the one number that survived the whole line. This lecture changes the scale: it looks at the **blocks** of a matrix rather than at its entries. Once a large matrix is cut into blocks, addition, scalar multiplication and multiplication all proceed block by block, and inversion reduces to two inverses of smaller matrices and two multiplications. The LU factorization of the next section is the same idea applied to row reduction: collect the multipliers used during elimination into one lower triangular matrix, and what comes out is a decomposition that can be reused.

Definition 1.1 (Partitioned matrix; Lay §2.4, p. 117)**.** Let $A \in M_{m\times n}(\mathbb{F})$. Draw several horizontal and vertical lines through A; each piece so obtained is a **submatrix** of A, the arrangement is a **partition** of A, and A itself is called a **partitioned matrix**. If the rows are cut into p groups and the columns into q groups, write $A = [A_{ij}]$, where A_{ij} is the submatrix lying at the i-th row group and the j-th column group.

Addition and **scalar multiplication** of partitioned matrices are performed block by block: provided the two matrices have the same size and are partitioned in exactly the same way, every block of $A + B$ is the sum of the corresponding blocks, and every block of cA is cA_{ij}.

Definition 1.2 (Conformable partitions; Lay §2.4, p. 118)**.** Let A be $m \times n$ and B be $n \times k$. If the **column** partition of A and the **row** partition of B break at the same points, the two partitions are **conformable for block multiplication**.

Theorem 1.1 (Block multiplication; Lay §2.4, p. 118)**.** *Let A be $m \times n$ and B be $n \times k$ with conformable partitions: the rows of A are cut into $m_1, \dots, m_p$ groups, the columns of A and the rows of B are both cut into $n_1, \dots, n_q$ groups, and the columns of B are cut into $k_1, \dots, k_r$ groups. Then AB, partitioned in the same way, is $[\,(AB)_{ij}\,]$ with*

$$(AB)_{ij} = \sum_{l=1}^{q} A_{il}B_{lj} \qquad (i = 1, \dots, p;\ j = 1, \dots, r).$$

Proof. Compare entries. The (x, y)-entry of AB is $\sum_{t=1}^{n} a_{xt}b_{ty}$. Group the index t by column groups: the l-th group pairs row x of A_{il} with column y of B_{lj} and adds the products, which is exactly the (x, y)-entry of $A_{il}B_{lj}$. Summing over the q groups gives the (x, y)-entry of $\sum_l A_{il}B_{lj}$, term by term equal to $\sum_t a_{xt}b_{ty}$. □

Theorem 1.2 (Column–row expansion; Lay Theorem 10, §2.4, p. 119)**.** *Let A be $m \times n$ and B be $n \times k$, and let $\mathrm{col}_k(A)$ be the k-th column of A ($m \times 1$) and $\mathrm{row}_k(B)$ the k-th row of B ($1 \times k$). Then*

$$AB = \sum_{k=1}^{n} \mathrm{col}_k(A)\ \mathrm{row}_k(B) = \begin{bmatrix} \mathrm{col}_1(A) & \cdots & \mathrm{col}_n(A) \end{bmatrix} \begin{bmatrix} \mathrm{row}_1(B) \\ \vdots \\ \mathrm{row}_n(B) \end{bmatrix}$$

Proof. The right-hand side is the block product for the partition of A into single columns and the partition of B into single rows, and those partitions are conformable. By Theorem 1.1 its (i, j)-entry is $\sum_k a_{ik}b_{kj}$, which is the (i, j)-entry of AB. □

Each block $\mathrm{col}_k(A)\,\mathrm{row}_k(B)$ in the brackets is an $m \times k$ matrix, a product of one column and one row (an **outer product**); the theorem writes AB as a sum of n matrices of rank at most 1. This is the fifth reading of AB; the other four are $A\boldsymbol{x}$ computed from the columns of A, the columns of AB computed from the columns of B, the entry-by-entry row-column rule, and the rows of AB computed from the rows of A; all five are catalogued in Lay §2.4, p. 118.

Definition 1.3 (Block upper triangular and block diagonal; Lay §2.4, p. 119–120)**.** Let A be a partitioned matrix. If every block below the main diagonal is a zero matrix, that is, if A has the form

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix},$$

then A is a **block upper triangular matrix**; if every block off the main diagonal is a zero matrix, then A is a **block diagonal matrix**.

Proposition 1.3 (Inverses of block triangular matrices; Lay Example 5, §2.4, p. 119–120)**.** *Let*

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

be block upper triangular, with A_{11} of size $p \times p$ and A_{22} of size $q \times q$. Then A is invertible if and only if both A_{11} and A_{22} are invertible, and in that case

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

In particular a block diagonal matrix is invertible if and only if every diagonal block is invertible, and its inverse is the block diagonal matrix of the inverses.

Proof. Suppose first that A_{11} and A_{22} are invertible and call the matrix on the right B. Compute AB and BA block by block: the upper-left block is $A_{11}A_{11}^{-1} = I_p$, the lower-right block is $A_{22}A_{22}^{-1} = I_q$, the lower-left block is $0 \cdot A_{11}^{-1} + A_{22} \cdot 0 = 0$, and the upper-right block is $A_{11}\big(-A_{11}^{-1}A_{12}A_{22}^{-1}\big) + A_{12}A_{22}^{-1} = -A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} = 0$; the four blocks of BA are the same. Hence $B = A^{-1}$.

Suppose now that A is invertible. Partition A^{-1} in the same way into four blocks B_{ij} (with B_{11} of size $p \times p$); from $AA^{-1} = I_{p+q}$ come the four block equations

$$A_{11}B_{11} + A_{12}B_{21} = I_p, \quad A_{11}B_{12} + A_{12}B_{22} = 0, \quad A_{22}B_{21} = 0, \quad A_{22}B_{22} = I_q.$$

The last one, together with "a square matrix with a one-sided inverse is invertible" (Lecture 4 notes, Theorem 3.7), makes A_{22} invertible with $B_{22} = A_{22}^{-1}$. The third then gives $B_{21} = A_{22}^{-1} \cdot 0 = 0$, so the first becomes $A_{11}B_{11} = I_p$, and the same argument makes A_{11} invertible with $B_{11} = A_{11}^{-1}$. Finally the second equation reads $A_{11}B_{12} = -A_{12}A_{22}^{-1}$, and left multiplication by A_{11}^{-1} gives $B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$. □

Example 1.1 (Block multiplication and a block inverse; written for these notes; method from Lay §2.4, p. 117–120)**.** Let

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

partitioned into 2×2 blocks (the first two rows and columns, then the last two). Find A^{-1} block by block, and check the answer by multiplying the two 4×4 matrices with the same partition.

Solution. With the $2 + 2$ partition,

$$A_{11} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2, \quad A_{21} = 0, \quad A_{22} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

The two diagonal blocks have determinants $2 \cdot 1 - 1 \cdot 1 = 1$ and $3 \cdot 1 - 1 \cdot 1 = 2$, both nonzero, so

$$A_{11}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad A_{22}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

The upper-right block is $-A_{11}^{-1}A_{12}A_{22}^{-1} = -A_{11}^{-1}A_{22}^{-1}$; compute the product first,

$$A_{11}^{-1}A_{22}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & -4 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -\frac{3}{2} & \frac{7}{2} \end{bmatrix}$$

and change its sign:

$$A^{-1} = \left[\begin{array}{cc|cc} 1 & -1 & -1 & 2 \\ -1 & 2 & \frac{3}{2} & -\frac{7}{2} \\ \hline 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{3}{2} \end{array}\right]$$

The check is done block by block: the upper-left block is $A_{11}A_{11}^{-1} + A_{12} \cdot 0 = I_2$; the lower-right block is $A_{22}A_{22}^{-1} = I_2$; the upper-right block is $A_{11}\big(-A_{11}^{-1}A_{22}^{-1}\big) + A_{12}A_{22}^{-1} = -A_{22}^{-1} + A_{22}^{-1} = 0$; the lower-left block is $0 \cdot A_{11}^{-1} + A_{22} \cdot 0 = 0$. All four are right, so $AA^{-1} = I_4$. □

Approach. The lower-left block is zero, so A is block upper triangular and Proposition 1.3 writes the inverse down: two 2×2 inverses and two 2×2 products suffice. Seeing a whole block of zeros, partition first; a 4×4 problem then turns into three 2×2 problems, which is half the work of row reducing this matrix directly.

2 LU Factorization

Row reduction is the standard move for solving a system, and it has one drawback: every new right-hand side $\boldsymbol{b}$ forces the whole elimination again. Recording the **multipliers** used along the way produces a pair of triangular matrices L and U, after which every new right-hand side costs two substitutions. That is what a **matrix factorization** means: writing a matrix as a product of simpler matrices. Lay §2.5 (p. 123) calls it an analysis of the data, matching the synthesis that multiplication performs.

Definition 2.1 (LU factorization; Lay §2.5, p. 124)**.** Let A be an $m \times n$ matrix. If there is an $m \times m$ **unit lower triangular** matrix L (lower triangular with every diagonal entry equal to 1) and an echelon form U of A such that

$$A = LU,$$

then this is an **LU factorization** of A.

Here U is only required to be *an* echelon form of A: it need not be square and need not have a pivot in every column. When A is $m \times n$, so is U, and its zero rows and its columns without a pivot stay where they are.

Theorem 2.1 (Row replacements produce an LU factorization; Lay §2.5, p. 125–126)**.** *Suppose A can be reduced to an echelon form U using only **row replacements**, that is, adding a multiple of one row to a row below it, and let $E_1, \dots, E_p$ be the elementary matrices of those replacements. Then*

$$L = (E_p \cdots E_1)^{-1}$$

is unit lower triangular, L is invertible, and $A = LU$.

Proof. Each row replacement is left multiplication by an elementary matrix (Lecture 5 notes, Theorem 3.1; Friedberg Theorem 3.1), so $E_p \cdots E_1 A = U$, that is, $A = (E_p \cdots E_1)^{-1}U$. The elementary matrix of a row replacement is itself unit lower triangular, and products and inverses of unit lower triangular matrices stay unit lower triangular (Lay §2.5, p. 126 and Exercise 19), so L is unit lower triangular; elementary matrices are invertible, so L is invertible. □

Proposition 2.2 (Solving a system with $A = LU$; Lay §2.5, p. 124)**.** *Let $A = LU$ be an LU factorization of A and let $\boldsymbol{b} \in \mathbb{F}^m$. Then $A\boldsymbol{x} = \boldsymbol{b}$ has the same solutions as the two triangular systems*

$$L\boldsymbol{y} = \boldsymbol{b}, \qquad U\boldsymbol{x} = \boldsymbol{y}.$$

The first is solved top-down (forward substitution), the second bottom-up (back substitution).

Proof. $A\boldsymbol{x} = L(U\boldsymbol{x})$. Put $\boldsymbol{y} = U\boldsymbol{x}$; then $A\boldsymbol{x} = \boldsymbol{b}$ is equivalent to "$L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$". Conversely, if $\boldsymbol{y}$ satisfies $L\boldsymbol{y} = \boldsymbol{b}$ and $\boldsymbol{x}$ satisfies $U\boldsymbol{x} = \boldsymbol{y}$, then $A\boldsymbol{x} = L\boldsymbol{y} = \boldsymbol{b}$. Since L is unit lower triangular, $L\boldsymbol{y} = \boldsymbol{b}$ has, for every right-hand side $\boldsymbol{b}$, a unique solution $\boldsymbol{y}$ found from the top (step i involves only y_i, with coefficient 1); hence L is invertible. □

Example 2.1 (Solving a system with $A = LU$; written for these notes; method from Lay §2.5, p. 124–125)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} = LU,$$

and solve $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{b} = (1, 0, 1)^\top$.

Solution. Forward substitution in $L\boldsymbol{y} = \boldsymbol{b}$:

$$y_1 = 1, \qquad y_2 = 0 - 2y_1 = -2, \qquad y_3 = 1 - y_1 - y_2 = 1 - 1 + 2 = 2.$$

Back substitution in $U\boldsymbol{x} = \boldsymbol{y}$: $2x_3 = 2$ gives $x_3 = 1$; $x_2 + x_3 = -2$ gives $x_2 = -3$; and $x_1 + 2x_2 + x_3 = 1$ gives $x_1 = 1 + 6 - 1 = 6$. Hence

$$\boldsymbol{x} = (6, -3, 1)^\top.$$

Check: $A\boldsymbol{x} = (6-6+1,\ 12-15+3,\ 6-9+4)^\top = (1, 0, 1)^\top$. The factorization stated in the problem can also be checked with one block product: the second row of LU is $2 \cdot (1, 2, 1) + (0, 1, 1) = (2, 5, 3)$ and the third is $(1, 2, 1) + (0, 1, 1) + (0, 0, 2) = (1, 3, 4)$. □

Approach. With L and U in hand, A is never touched again: solve $L\boldsymbol{y} = \boldsymbol{b}$ first (the first row involves only y_1, the second only y_1, y_2, so substitute downwards), then $U\boldsymbol{x} = \boldsymbol{y}$ (start from the last row and work up). Both triangular systems are pure substitution; no elimination is left.

Remark 2.1 (The LU algorithm; Lay §2.5, p. 126)**.**

1. Reduce A to an echelon form U by row replacements, recording each **multiplier** (the c in "row i plus c times row j", subtracting being the case of negative c).
2. Put the opposite of each multiplier into position (i, j) of L with $i > j$, put 1 on the diagonal of L, and 0 everywhere else.

Step 2 rests on the definition $L = (E_p \cdots E_1)^{-1}$: left multiplication by $E_p \cdots E_1$ is what carries L back to I, so the very sequence of row replacements that carries A to U carries L to I, and the (i, j)-entry of L is the multiplier used to clear row i of column j, with the sign changed. One multiplication verifies the pair: LU should reproduce A.

Example 2.2 (An LU factorization of a 4×5 matrix; Lay Example 2, §2.5, p. 126–127)**.** Find an LU factorization of

$$A = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$$

Solution. First clear column 1 with row 1: add twice row 1 to row 2, subtract row 1 from row 3, and add three times row 1 to row 4:

$$\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & -9 & -3 & -4 & 10 \\ 0 & 12 & 4 & 12 & -5 \end{bmatrix}$$

Then clear column 2 with row 2: add three times row 2 to row 3 and subtract four times row 2 from row 4:

$$\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 & 7 \end{bmatrix}$$

Column 3 has no pivot, so move to column 4: subtract twice row 3 from row 4, and the echelon form is

$$U = \begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ 0 & 3 & 1 & 2 & -3 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

Recorded as the added multiple c, the multipliers are $c = 2, -1, 3$ (column 1, rows $2, 3, 4$), then $c = 3, -4$ (column 2, rows $3, 4$), then $c = -2$ (column 4, row 4); with the sign changed they fill L:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -3 & 1 & 0 \\ -3 & 4 & 2 & 1 \end{bmatrix}$$

Check: the second row of LU is $2 \cdot (2, 4, -1, 5, -2) + (0, 3, 1, 2, -3) = (-4, -5, 3, -8, 1)$, which is row 2 of A; the other three rows match the same way. □

Approach. Eliminate column by column from the left, and after clearing each column put the multipliers with the sign changed into the corresponding column of L; U is the echelon form that results. A has four rows, so L is 4×4. If some column has no pivot, skip it and keep that pivotless column in U.

Remark 2.2 (Permuted LU and partial pivoting; Lay §2.5, p. 127)**.** In practice row interchanges are almost always needed: partial pivoting takes, as the pivot, the entry of largest absolute value in the column, so that no division by a tiny number occurs. Once interchanges are allowed, L is no longer lower triangular but **permuted lower triangular**: rearranging the rows of L in the order the pivots appear turns it back into a unit lower triangular matrix. Solving works as before, except that forward substitution follows the pivot order from left to right. The phrase "LU factorization" usually includes this permuted case.

Remark 2.3 (Why LU rather than A^{-1}; Lay §2.5, p. 127)**.** An LU factorization costs about $2n^3/3$ multiplications and additions, the same order as one elimination; afterwards each new right-hand side costs $O(n^2)$ for the two substitutions. Solving through A^{-1} first pays $O(n^3)$ to produce A^{-1}, and when A is sparse A^{-1} is usually dense, so multiplying by it is slow. For a sequence of right-hand sides with the same A, the difference is the difference between "factor once, substitute many times" and "invert again for every right-hand side".

3 Computing the Inverse

Lecture 4 established that an $n \times n$ matrix is invertible if and only if it is row equivalent to I_n. That theorem answers yes or no; this section opens its proof to read off an algorithm: attach I_n to the right of A, row reduce the whole array, and when the left half has become I_n the right half has become A^{-1}.

Definition 3.1 (Augmented matrix; Friedberg §3.2, p. 161)**.** Let A be $m \times n$ and B be $m \times p$. The $m \times (n + p)$ matrix obtained by placing B to the right of A is written $(A \mid B)$ and called the **augmented matrix** of A and B: its first n columns are the columns of A, its last p columns the columns of B.

Theorem 3.1 (An inverse by row reduction; Friedberg §3.2, p. 161–162)**.** *Let A be an $n \times n$ matrix.*

(1) If A is invertible, then elementary row operations carry $(A \mid I_n)$ to $(I_n \mid A^{-1})$.

(2) Conversely, if some sequence of elementary row operations carries $(A \mid I_n)$ to $(I_n \mid B)$, then A is invertible and $B = A^{-1}$.

Proof. (1) By Lecture 5 notes, Proposition 3.8 (Friedberg Corollary 3, p. 159), an invertible matrix is a product of elementary matrices; write $A^{-1} = E_p \cdots E_1$. Left multiplication by an elementary matrix is an elementary row operation, so

$$E_p \cdots E_1 (A \,|\, I_n) = (A^{-1}A \,|\, A^{-1}I_n) = (I_n \,|\, A^{-1}),$$

that is, the sequence carries $(A \,|\, I_n)$ to $(I_n \,|\, A^{-1})$.

(2) Let $E_1, \ldots, E_p$ be the elementary matrices of that sequence and put $M = E_p \cdots E_1$. Then $M(A \,|\, I_n) = (MA \,|\, M) = (I_n \,|\, B)$, so comparing the two halves gives $MA = I_n$ and $M = B$. By "a square matrix with a one-sided inverse is invertible" (Lecture 4 notes, Theorem 3.7), A is invertible with $M = A^{-1}$, hence $B = A^{-1}$. □

Remark 3.1 (What failure looks like; Friedberg §3.2, p. 162)**.** If A is not invertible, every attempt to carry $(A \,|\, I_n)$ to $(I_n \,|\, B)$ must fail: elementary row operations preserve rank (Lecture 5 notes, Theorem 3.4), and $\operatorname{rank}(A) < n$, so the left half produces at most $n-1$ pivots and some row whose first n entries are all zero must appear. Once such a row shows up there is no point continuing: A is not invertible.

Example 3.1 (An inverse by row reduction; Friedberg Example 5, §3.2, p. 162–163)**.** Decide whether

$$A = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix}$$

is invertible, and if it is, find A^{-1}.

Solution.

$$\left[\begin{array}{ccc|ccc} 0 & 2 & 4 & 1 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 2 & 4 & 2 & 0 & 1 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & \frac{1}{2} & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 0 & -3 & -2 & 0 & -\frac{3}{2} & 1 \end{array}\right]$$

(interchange rows 1 and 2; multiply row 1 by $\frac{1}{2}$; subtract three times row 1 from row 3). Then

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & -3 & -2 & 0 & -\frac{3}{2} & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 4 & \frac{3}{2} & -\frac{3}{2} & 1 \end{array}\right]$$

(multiply row 2 by $\frac{1}{2}$; subtract twice row 2 from row 1; add three times row 2 to row 3). Finally

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{array}\right]$$

(multiply row 3 by $\frac{1}{4}$; add three times row 3 to row 1; subtract twice row 3 from row 2). The left half is I_3, so A is invertible and

$$A^{-1} = \frac{1}{8} \begin{bmatrix} 1 & -5 & 6 \\ -2 & 6 & -4 \\ 3 & -3 & 2 \end{bmatrix}$$

Check: row 1 of A times column 1 of A^{-1} is $\big(0 \cdot 1 + 2 \cdot (-2) + 4 \cdot 3\big)/8 = 8/8 = 1$, and row 1 times column 2 is $\big(0 \cdot (-5) + 2 \cdot 6 + 4 \cdot (-3)\big)/8 = 0/8 = 0$. □

Approach. Attach I_3 on the right and row reduce only the left half, letting the right half move along. The first entry is 0, so interchange rows 1 and 2; then clear each column in turn, scaling the pivot to 1 and eliminating the rest of its column. When the left half has become I_3, the right half is A^{-1}; if a whole zero row appears on the left, A is not invertible.

Remark 3.2 (A whole A^{-1} from one LU; Lay §2.5, p. 127)**.** The j-th column of A^{-1} is the solution of $A\boldsymbol{x} = \boldsymbol{e}_j$. So once $A = LU$ is known, computing the whole inverse means taking $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ as right-hand sides in turn, each with one forward and one back substitution: factor once, substitute n times. Compared with "compute A^{-1} and multiply", what is saved is the n-th elimination.

4 Solution Sets of Linear Systems

The previous section answered "when does $A\boldsymbol{x} = \boldsymbol{b}$ have exactly one solution", and it used the inverse of an $n \times n$ matrix to do it. Widening the same question to an arbitrary $m \times n$ A starts with the shape of the solution set itself. The result of this section fits in one sentence: **the solution set of a homogeneous system is the kernel of L_A, and the solution set of a nonhomogeneous system is that kernel shifted by one particular solution**. Lay describes the same thing in §1.5 with free variables; here that description is rewritten in the language of subspaces.

Definition 4.1 (Homogeneous and nonhomogeneous systems; Friedberg §3.3, p. 171)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and $\boldsymbol{b} \in \mathbb{F}^m$. The system $A\boldsymbol{x} = \boldsymbol{b}$ is **homogeneous** when $\boldsymbol{b} = \boldsymbol{0}$ and **nonhomogeneous** when $\boldsymbol{b} \neq \boldsymbol{0}$. The matrix A defines the left-multiplication map $L_A : \mathbb{F}^n \to \mathbb{F}^m$, $L_A(\boldsymbol{x}) = A\boldsymbol{x}$, so that

$$A\boldsymbol{x} = \boldsymbol{b} \iff L_A(\boldsymbol{x}) = \boldsymbol{b}, \qquad A\boldsymbol{x} = \boldsymbol{0} \iff L_A(\boldsymbol{x}) = \boldsymbol{0}.$$

A homogeneous system always has the solution $\boldsymbol{x} = \boldsymbol{0}$. Write K for the solution set of $A\boldsymbol{x} = \boldsymbol{b}$ and K_H for the solution set of the corresponding homogeneous system.

Theorem 4.1 (A homogeneous solution set is the kernel; Friedberg Theorem 3.8, §3.3, p. 171)**.** *Let $A\boldsymbol{x} = \boldsymbol{0}$ be a homogeneous system of m equations in n unknowns and let K be its solution set. Then*

$$K = \mathcal{N}(L_A),$$

so K is a subspace of $\mathbb{F}^n$, and

$$\dim K = n - \operatorname{rank}(A).$$

Proof. $\boldsymbol{x} \in K$ if and only if $A\boldsymbol{x} = \boldsymbol{0}$, that is $L_A(\boldsymbol{x}) = \boldsymbol{0}$, that is $\boldsymbol{x} \in \mathcal{N}(L_A)$; the kernel of a matrix transformation is the null space of A (Lecture 3 notes, Proposition 2.2), and a kernel is a subspace (Lecture 3 notes, Theorem 2.1). The dimension follows from the Dimension Theorem (Lecture 3 notes, Theorem 2.4) together with $\operatorname{rank}(A) = \dim \mathcal{R}(L_A)$. □

Proposition 4.2 (More unknowns than equations forces a nonzero solution; Friedberg §3.3, p. 171, Corollary)**.** *Let A be an $m \times n$ matrix. If $m < n$, then $A\boldsymbol{x} = \boldsymbol{0}$ has a nonzero solution.*

Proof. $\operatorname{rank}(A) \leq m$, so by Theorem 4.1, $\dim K = n - \operatorname{rank}(A) \geq n - m > 0$, hence $K \neq \{\boldsymbol{0}\}$. □

Theorem 4.3 (A nonhomogeneous solution set is a shift of the kernel; Friedberg Theorem 3.9, §3.3, p. 172)**.** *Let K be the solution set of $A\boldsymbol{x} = \boldsymbol{b}$, let K_H be the solution set of the corresponding system $A\boldsymbol{x} = \boldsymbol{0}$, and let $\boldsymbol{s}$ be any solution of $A\boldsymbol{x} = \boldsymbol{b}$. Then*

$$K = \{\boldsymbol{s}\} + K_H = \{\boldsymbol{s} + \boldsymbol{k} : \boldsymbol{k} \in K_H\}.$$

Proof. First $K \subseteq \{\boldsymbol{s}\} + K_H$. Take $\boldsymbol{w} \in K$; then $A\boldsymbol{w} = \boldsymbol{b} = A\boldsymbol{s}$, so $A(\boldsymbol{w} - \boldsymbol{s}) = \boldsymbol{0}$, that is $\boldsymbol{w} - \boldsymbol{s} \in K_H$, and $\boldsymbol{w} = \boldsymbol{s} + \boldsymbol{k}$ with $\boldsymbol{k} = \boldsymbol{w} - \boldsymbol{s}$. Conversely, if $\boldsymbol{w} = \boldsymbol{s} + \boldsymbol{k}$ with $\boldsymbol{k} \in K_H$, then $A\boldsymbol{w} = A\boldsymbol{s} + A\boldsymbol{k} = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$, so $\boldsymbol{w} \in K$. □

Remark 4.1 (How to read the two sentences)**.** Theorem 4.1 says the homogeneous solution set is a subspace **through the origin**; Theorem 4.3 says the nonhomogeneous solution set is that

subspace **shifted** onto a particular solution. How many solutions a nonhomogeneous system has is therefore decided entirely by K_H: $K_H = \{\mathbf{0}\}$ means a unique solution, and a nonzero vector in K_H means as many solutions as $\mathbb{F}$ has elements.

For an $n \times n$ matrix, $K_H = \{\mathbf{0}\}$ is exactly invertibility; the solution set is then the single point $\boldsymbol{x} = A^{-1}\boldsymbol{b}$, which is Lecture 4 notes, Theorem 3.2. The converse holds too: if $A\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution for every $\boldsymbol{b}$, then A is invertible (Friedberg Theorem 3.10, §3.3, p. 174).

Example 4.1 (Solving a system with the inverse of the previous section; Friedberg Example 4, §3.3, p. 174)**.** Solve the system

$$2x_2 + 4x_3 = 2, \qquad 2x_1 + 4x_2 + 2x_3 = 3, \qquad 3x_1 + 3x_2 + x_3 = 1;$$

its coefficient matrix is the A whose inverse was computed in Example 3.1.

Solution. Write the system as $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{b} = (2, 3, 1)^\top$, and recall

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 1 & -5 & 6 \\ -2 & 6 & -4 \\ 3 & -3 & 2 \end{bmatrix}.$$

Then

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = \frac{1}{8}\begin{bmatrix} 1\cdot 2 - 5\cdot 3 + 6\cdot 1 \\ -2\cdot 2 + 6\cdot 3 - 4\cdot 1 \\ 3\cdot 2 - 3\cdot 3 + 2\cdot 1 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -7 \\ 10 \\ -1 \end{bmatrix} = \left(-\tfrac{7}{8},\ \tfrac{5}{4},\ -\tfrac{1}{8}\right)^\top.$$

Check: the three entries of $A\boldsymbol{x}$ are $2\cdot\frac{5}{4} + 4\cdot\left(-\frac{1}{8}\right) = 2$, $2\cdot\left(-\frac{7}{8}\right) + 4\cdot\frac{5}{4} + 2\cdot\left(-\frac{1}{8}\right) = 3$ and $3\cdot\left(-\frac{7}{8}\right) + 3\cdot\frac{5}{4} + \left(-\frac{1}{8}\right) = 1$, matching the right-hand side. □

Approach. Look at the coefficient matrix first. Its determinant is nonzero and the previous section already produced A^{-1}, so by Lecture 4 notes, Theorem 3.2 the system has the unique solution $\boldsymbol{x} = A^{-1}\boldsymbol{b}$: multiply the inverse by the right-hand side and no second row reduction is needed. This is the case $K_H = \{\mathbf{0}\}$ of Theorem 4.3.

5 Reading a Solution Set off the Reduced Echelon Form

The previous section wrote the solution set as $\{\boldsymbol{s}\} + K_H$ but did not say where $\boldsymbol{s}$ and a basis of K_H come from. This section supplies the procedure: reduce the augmented matrix to **reduced row echelon form**; the pivot columns give the basic variables and the remaining columns the free variables; setting every free variable to 0 gives a particular solution, and setting one of them to 1 at a time gives a basis of K_H. The number of vectors obtained is $n - r$, exactly the dimension the Dimension Theorem predicts.

Definition 5.1 (Reduced row echelon form; Friedberg §3.4, p. 185)**.** A matrix is in **reduced row echelon form** if all three of the following hold:

(a) every row containing a nonzero entry precedes every row that is entirely zero;
(b) the first nonzero entry of each row is the only nonzero entry of its column;
(c) the first nonzero entry of each row equals 1 and lies in a column to the right of the first nonzero entry of the preceding row.

Theorem 5.1 (Invertible left multiplication preserves the solution set; Friedberg Theorem 3.13, §3.4, p. 182)**.** *Let $A\boldsymbol{x} = \boldsymbol{b}$ be a system of m equations in n unknowns and let C be an invertible $m \times m$ matrix. Then $CA\boldsymbol{x} = C\boldsymbol{b}$ and $A\boldsymbol{x} = \boldsymbol{b}$ have the same solutions.*

Proof. Let K and K' be the two solution sets. If $\boldsymbol{w} \in K$, then $A\boldsymbol{w} = \boldsymbol{b}$, and left multiplication by C gives $CA\boldsymbol{w} = C\boldsymbol{b}$, so $\boldsymbol{w} \in K'$. Conversely, if $\boldsymbol{w} \in K'$, then $CA\boldsymbol{w} = C\boldsymbol{b}$; left multiplication by C^{-1} gives $A\boldsymbol{w} = C^{-1}C\boldsymbol{b} = \boldsymbol{b}$, so $\boldsymbol{w} \in K$. □

Proposition 5.2 (Elementary row operations preserve the solution set; Friedberg §3.4, p. 182, Corollary)**.** *If* $(\,A' \mid \boldsymbol{b}'\,)$ *is obtained from* $(\,A \mid \boldsymbol{b}\,)$ *by finitely many elementary row operations, then* $A'\boldsymbol{x} = \boldsymbol{b}'$ *and* $A\boldsymbol{x} = \boldsymbol{b}$ *have the same solutions.*

Proof. Each elementary row operation is left multiplication by an elementary matrix (Lecture 5 notes, Theorem 3.1), and elementary matrices are invertible (Lecture 5 notes, Theorem 3.2), so each step leaves the solution set unchanged by Theorem 5.1; finitely many such steps leave it unchanged as well. □

Remark 5.1 (The two steps that read off a solution set)**.** Let $(\,A \mid \boldsymbol{b}\,)$ be in reduced row echelon form, with pivots in columns $j_1 < \cdots < j_r$; the other $n - r$ columns $f_1 < \cdots < f_{n-r}$ are the free ones. Write β_i for the entry of the i-th nonzero row in the augmented column and c_{ik} for its entry in column f_k. That row then reads

$$x_{j_i} + c_{i1}x_{f_1} + \cdots + c_{i,n-r}x_{f_{n-r}} = \beta_i,$$

that is,

$$x_{j_i} = \beta_i - c_{i1}x_{f_1} - \cdots - c_{i,n-r}x_{f_{n-r}}.$$

Step one: treat the free variables as parameters, $x_{f_k} = t_k$; the displayed identity gives each basic variable in turn. Step two: collect the result by parameter to obtain

$$\boldsymbol{s} = \boldsymbol{s}_0 + t_1\boldsymbol{u}_1 + \cdots + t_{n-r}\boldsymbol{u}_{n-r},$$

where $\boldsymbol{s}_0$ carries β_i in the pivot positions and 0 in the free positions, while $\boldsymbol{u}_k$ carries 1 in position f_k, 0 in the other free positions, and $-c_{ik}$ in the pivot positions.

This table exhausts the solution set: any choice of $t_1, \ldots, t_{n-r}$ satisfies every row and so is a solution, and conversely, once the free components of a solution are fixed, each pivot component is determined by its own row, leaving nothing else.

Theorem 5.3 (Reading a solution set off the reduced echelon form; Friedberg Theorem 3.15, §3.4, p. 189–190)**.** *Let* $A\boldsymbol{x} = \boldsymbol{b}$ *be a system of* r *nonzero equations in* n *unknowns with* $\operatorname{rank}(A) = \operatorname{rank}(A \,|\, \boldsymbol{b})$*, and suppose* $(\,A \mid \boldsymbol{b}\,)$ *is in reduced row echelon form. Then*

(a) $\operatorname{rank}(A) = r$*;*

(b) if the general solution obtained by the two steps of Remark 5.1 is written as $\boldsymbol{s} = \boldsymbol{s}_0 + t_1\boldsymbol{u}_1 + \cdots + t_{n-r}\boldsymbol{u}_{n-r}$*, then* $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$ *is a basis of* K_H *and* $\boldsymbol{s}_0$ *is a solution of the original system.*

Proof. (a) A matrix in reduced row echelon form has exactly r nonzero rows, and by definition those rows are linearly independent, so $\operatorname{rank}(A \,|\, \boldsymbol{b}) = r$, hence $\operatorname{rank}(A) = r$.

(b) Setting $t_1 = \cdots = t_{n-r} = 0$ gives $\boldsymbol{s} = \boldsymbol{s}_0$, so $\boldsymbol{s}_0$ solves the original system. By Theorem 4.3, $K = \{\boldsymbol{s}_0\} + K_H$, hence $K_H = \{-\boldsymbol{s}_0\} + K = \operatorname{Span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$. Also $\dim K_H = n - \operatorname{rank}(A) = n - r$ (Theorem 4.1), and the spanning set has only $n - r$ vectors, so it is a basis of K_H. □

Example 5.1 (Reading a general solution off the reduced echelon form; classroom example of 30 October 2026)**.** The augmented matrix of a system has already been reduced to

$$\left[\begin{array}{ccccc|c} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Write down its general solution and a basis of the solution set K_H of the corresponding homogeneous system.

Solution. The pivots lie in columns $1, 2, 5$, so x_1, x_2, x_5 are the **basic variables** and x_3, x_4 the **free variables**; there are 3 nonzero rows, so $r = 3$ and there are $n - r = 2$ free variables.

Solve each basic variable from the equation of its own row:

$$x_1 = -24 + 2x_3 - 3x_4, \qquad x_2 = -7 + 2x_3 - 2x_4, \qquad x_5 = 4.$$

Treating x_3, x_4 as parameters t_1, t_2, the general solution is

$$\boldsymbol{s} = \begin{bmatrix} -24 \\ -7 \\ 0 \\ 0 \\ 4 \end{bmatrix} + t_1 \begin{bmatrix} 2 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -3 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \boldsymbol{s}_0 + t_1 \boldsymbol{u}_1 + t_2 \boldsymbol{u}_2,$$

where $\boldsymbol{s}_0$ is the particular solution at $t_1 = t_2 = 0$ and $\boldsymbol{u}_1, \boldsymbol{u}_2$ solve the homogeneous system. Check: each row of $A\boldsymbol{u}_1$ and $A\boldsymbol{u}_2$ is "nonzero entry times the corresponding component, summed", and the three rows give $2 - 2 = 0$, $2 - 2 = 0$ and 0, so both vectors lie in K_H; the three rows of $A\boldsymbol{s}_0$ are -24, -7 and 4, matching the augmented column.

The vectors $\boldsymbol{u}_1, \boldsymbol{u}_2$ are linearly independent: their 3rd and 4th components are $(1, 0)$ and $(0, 1)$, so $c_1\boldsymbol{u}_1 + c_2\boldsymbol{u}_2 = \boldsymbol{0}$ forces $c_1 = c_2 = 0$ on those two components. Since $\dim K_H = n - r = 2$, the pair $\{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ is a basis of K_H, and every solution can be written as

$$K = \{\boldsymbol{s}_0\} + \operatorname{Span}(\boldsymbol{u}_1, \boldsymbol{u}_2) = \{\boldsymbol{s}_0\} + K_H,$$

which is the concrete form of Theorem 4.3. □

Approach. Faced with a reduced row echelon form, count the pivots to name the basic variables; the remaining columns are the free ones. Treat the free variables as parameters and solve each row for its own basic variable. The particular solution sets every parameter to zero; the basis vectors set one parameter to 1 and the rest to zero. The $n - r$ vectors obtained this way are automatically independent, because their free-variable components are the standard basis vectors, and $\dim K_H = n - r$, so there is nothing left to check about spanning.

6 A Worked Example

Example 6.1 (Factor, solve and invert one column; written for these notes; methods from Lay §2.5, p. 123–127 and Friedberg §3.2, p. 161–163)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 3 & 1 \\ -1 & -1 & 5 & -1 \\ 3 & 4 & -6 & 8 \end{bmatrix}, \qquad \boldsymbol{b} = (0, 4, 9, -5)^{\top}.$$

(1) Reduce A to an echelon form U using row replacements only, and write down L with $A = LU$. (2) Solve $A\boldsymbol{x} = \boldsymbol{b}$ with that one pair L, U. (3) Using the same L, U and nothing else, find column 2 of A^{-1}. (4) Decide whether A is invertible and state $\operatorname{rank}(A)$.

Solution. **(1) The factorization.** Subtract twice row 1 from row 2, add row 1 to row 3, subtract three times row 1 from row 4:

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 1 & 5 & 0 \\ 0 & -2 & -6 & 5 \end{bmatrix}$$

then subtract row 2 from row 3 and add twice row 2 to row 4:

$$U = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

The multipliers are $2, -1, 3$ (column 1, rows $2, 3, 4$), then $1, -2$ (column 2, rows $3, 4$), then 0 (column 3, row 4); with signs changed they fill L:

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 3 & -2 & 0 & 1 \end{bmatrix}$$

Check: row 3 of LU is $(-1) \cdot (1, 2, 0, 1) + (0, 1, 3, -1) + (0, 0, 2, 1) = (-1, -1, 5, -1)$ and row 4 is $3 \cdot (1, 2, 0, 1) - 2 \cdot (0, 1, 3, -1) + (0, 0, 0, 3) = (3, 4, -6, 8)$, both matching A.

(2) Solving $A\boldsymbol{x} = \boldsymbol{b}$. Forward substitution in $L\boldsymbol{y} = \boldsymbol{b}$:

$$y_1 = 0, \qquad y_2 = 4 - 2 \cdot 0 = 4,$$
$$y_3 = 9 - \big(-y_1 + y_2\big) = 9 - 4 = 5, \qquad y_4 = -5 - \big(3y_1 - 2y_2 + 0 \cdot y_3\big) = -5 + 8 = 3.$$

Back substitution in $U\boldsymbol{x} = \boldsymbol{y}$: $3x_4 = 3$ gives $x_4 = 1$; $2x_3 + x_4 = 5$ gives $x_3 = 2$; $x_2 + 3x_3 - x_4 = 4$ gives $x_2 = 4 - 6 + 1 = -1$; $x_1 + 2x_2 + x_4 = 0$ gives $x_1 = 1$. Hence

$$\boldsymbol{x} = (1, -1, 2, 1)^\top.$$

Check: $A\boldsymbol{x} = (1 - 2 + 0 + 1,\ 2 - 5 + 6 + 1,\ -1 + 1 + 10 - 1,\ 3 - 4 - 12 + 8)^\top = (0, 4, 9, -5)^\top$.

(3) Column 2 of A^{-1}. That column solves $A\boldsymbol{x} = \boldsymbol{e}_2$, so the right-hand side becomes $(0, 1, 0, 0)^\top$ with L, U unchanged. Forward: $y_1 = 0$, $y_2 = 1 - 0 = 1$, $y_3 = 0 - \big(-y_1 + y_2\big) = -1$, $y_4 = 0 - \big(3y_1 - 2y_2\big) = 2$. Back: $3x_4 = 2$ gives $x_4 = \frac{2}{3}$; $2x_3 + \frac{2}{3} = -1$ gives $x_3 = -\frac{5}{6}$; $x_2 + 3 \cdot \big(-\frac{5}{6}\big) - \frac{2}{3} = 1$ gives $x_2 = \frac{25}{6}$; and $x_1 + 2 \cdot \frac{25}{6} + \frac{2}{3} = 0$ gives $x_1 = -9$. Hence

$$\text{column 2 of } A^{-1} = \Big(-9,\ \tfrac{25}{6},\ -\tfrac{5}{6},\ \tfrac{2}{3}\Big)^\top.$$

Check: multiplying A by this column, the second row gives $2 \cdot (-9) + 5 \cdot \frac{25}{6} + 3 \cdot \big(-\frac{5}{6}\big) + 1 \cdot \frac{2}{3} = -18 + \frac{125}{6} - \frac{15}{6} + \frac{4}{6} = 1$, the second entry of $\boldsymbol{e}_2$; the other three rows give 0.

(4) Invertibility and rank. U has the four pivots $1, 1, 2, 3$, all nonzero, so no row was lost and A is row equivalent to I_4; by the Invertible Matrix Theorem (Lecture 4 notes, Theorem 3.6) A is invertible, and the number of pivots is the rank, so $\operatorname{rank}(A) = 4$. This agrees with Lecture 5: elementary row operations do not change the rank (Lecture 5 notes, Theorem 3.4). □

Approach. All four parts share one elimination. In (1) record the multiplier at each step and put its opposite into L; in (2) forward substitute for $L\boldsymbol{y} = \boldsymbol{b}$ and back substitute for $U\boldsymbol{x} = \boldsymbol{y}$; in (3) take $\boldsymbol{e}_2$ as the new right-hand side and repeat the same two substitutions without factoring again; in (4) every diagonal entry of U is nonzero, so all four pivots are present, the Invertible Matrix Theorem (Lecture 4 notes, Theorem 3.6) makes A invertible, and the rank is the number of pivots, 4.

7 Supplement: notions Homework 6 uses but the lecture did not cover

Remark 7.1 (Where these come from)**.** None of the following sits on the lecture's main line, yet Homework 6 uses it:

- the **annihilator** S^0: Homework sheet, Definition 1; Friedberg §2.6, the preamble to Exercises 13–17 (printed page 126);
- the **canonical map** ψ: the second half of Definition 1 on the sheet. It is the ψ of Lecture 5 notes, Theorem 2.4, that is $\psi(\boldsymbol{x})(f) = f(\boldsymbol{x})$;
- **the transpose of a transpose** $(T^\top)^\top$: Homework sheet, Problem 1(f); apply the transpose of Lecture 5 notes, Theorem 2.2 twice;
- **the sum of subsets** $W_1 + W_2$: Homework sheet, Problem 11(a); the definition is in the supplement of Lecture 2 notes (Friedberg §1.3 exercises, printed page 22).

Definition 7.1 (Annihilator; Homework sheet, Definition 1; Friedberg §2.6, preamble to Exercises 13–17, p. 126)**.** Let V be a finite-dimensional vector space and $S \subseteq V$. The **annihilator** of S is

$$S^0 = \{\, f \in V^* :\ f(\boldsymbol{x}) = 0 \text{ for all } \boldsymbol{x} \in S \,\}.$$

When W is a subspace of V, the set W^0 is called the annihilator of W.

8 Checklist and Exercises

By the end of this lecture you should be able to:

1. state what a partitioned matrix, a submatrix and a conformable pair of partitions are, and write $(AB)_{ij} = \sum_l A_{il}B_{lj}$ block by block;
2. expand AB into a sum of n outer products and explain why its (i, j)-entry is $\sum_k a_{ik}b_{kj}$;
3. state the criterion for a block upper triangular matrix to be invertible, write down its inverse, and point out which theorem the step "A invertible $\Rightarrow$ both diagonal blocks invertible" uses;
4. define an LU factorization and explain why U is only required to be an echelon form, not an upper triangular square matrix;
5. split $A\boldsymbol{x} = \boldsymbol{b}$ into $L\boldsymbol{y} = \boldsymbol{b}$ and $U\boldsymbol{x} = \boldsymbol{y}$ and solve each by triangular substitution;
6. build the L of a sequence of row replacements by recording the multipliers and changing their signs, and verify $LU = A$ with one multiplication;
7. compute the inverse of a 3×3 matrix through $(A \,|\, I) \to (I \,|\, A^{-1})$, and say what the left half must show when the matrix is not invertible;
8. obtain any column of A^{-1} from one LU factorization, and say which step that saves compared with "compute A^{-1} and then multiply";
9. tell a homogeneous system from a nonhomogeneous one, and state that its solution set is $\mathcal{N}(L_A)$ of dimension $n - \operatorname{rank}(A)$;
10. write the solution set of a nonhomogeneous system as a particular solution plus the homogeneous solution set, and decide when it is a single point;
11. read the basic and free variables off a reduced row echelon form, write down a particular solution and a basis of K_H, and say why those $n - r$ vectors are automatically independent.

Exercises (printed pages). Lay: §2.4 p. 121–122 (Exercises 1–16: block operations, block diagonal matrices, the Schur complement); §2.5 p. 129 (Exercises 1–6: solving with an LU factorization; 7–16: finding one); §2.2 p. 109 (Exercises 1–4 and 29–34: the algorithm for an inverse); §2.3 p. 115 (Exercises 1–10: deciding invertibility); §1.5 p. 47 (Exercises 1–16: solution sets of homogeneous and nonhomogeneous systems); §1.2 p. 21 (Exercises 1–14: echelon and reduced echelon forms); Friedberg: §3.2 p. 165 (Exercises 1–15: rank, inverses and augmented matrices); §3.3 p. 179 (Exercises 1–11: homogeneous solution sets and particular solutions); §3.4 p. 194 (Exercises 1–10: reduced echelon forms and general solutions). Each section is preceded by Practice Problems; do those first.

How to use the two texts together:

- **Computation and proof divide cleanly.** How a block product is computed, how L and U are filled in, how the row reduction for an inverse runs: Lay (§2.4, §2.5, §2.2). Why the

$(A \mid I)$ algorithm is legitimate, and why L must come out unit lower triangular: Friedberg (§3.2) together with Lay's account of the algorithm.

- The spine of the lecture: *partitions → block multiplication and the column–row expansion → the inverse of a block triangular matrix → collect the multipliers into L and get $A = LU$ → solve by forward and back substitution → substitute n times again for the columns of A^{-1}.*
- The one sentence worth keeping is $A = LU$: it turns elimination, a one-off action, into a reusable factorization. After one factorization, any number of right-hand sides and every column of A^{-1} cost nothing but substitution; the block section offers the same saving in another form, cutting the matrix so that products and inverses happen on smaller matrices.